

Aldo F. Galvan

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Summary

PhD graduate in Mechanical Engineering (Robotics) from UT Austin, specializing in bilateral teleoperation, dexterous manipulation, demonstration data pipelines, and vision-based perception. Led large-scale demonstration collection with Sony AI and deployed real-time perception systems on real hardware.

Education

PhD, Mechanical Engineering (Robotics), The University of Texas at Austin

May 2026

Advisor: Ann Majewicz Fey

B.S., Mechanical Engineering, University of Texas at Dallas

Technical Skills

Robotics & Control: Kinematics, bilateral teleoperation, optimization-based retargeting, haptic rendering, real-time low-latency control, system identification.

Perception: Stereo depth estimation, semantic segmentation, 6-DoF pose estimation, ICP-based registration, camera calibration, synthetic data generation with domain randomization, model fine-tuning.

Data & Pipelines: Multi-modal demonstration collection, time-synchronized data schemas, HDF5.

Frameworks & Languages: Python, C/C++, CUDA, ROS/ROS2, PyTorch, Pinocchio, OpenCV, Git, Linux.

Simulators & Hardware: Isaac Sim, Bullet Physics; Intuitive dVRK, Force Dimension Lambda.7, Roboligent Optimo, UT Austin Maestro exoskeleton, UT Austin Plato v2 hand.

Experience

Graduate Research Assistant, Human Enabled Robotic Technology (HERO) Lab, UT Austin Aug 2020 – May 2026

Research on bilateral teleoperation, dexterous manipulation, demonstration data pipelines, and vision-based perception across manipulation and surgical platforms.

Selected Projects

Large-Scale Demonstration Collection on Manipulator and Dexterous Hand

Sony AI collaboration: bilateral teleoperation stack and multimodal data pipeline

- Led large-scale human demonstration collection on a Roboligent Optimo manipulator with a UT Austin Plato v2 dexterous hand (Sony AI collaboration); partly owned the teleoperation infrastructure and data collection feeding collaborators' imitation learning.
- Developed the bilateral teleoperation stack: retargeting from a Force Dimension Lambda.7 haptic device, learning-based system identification, and hand-eye calibration; also supports a UT Austin Maestro wearable exoskeleton as an alternate input.
- Designed a time-aligned multi-modal data schema (proprioception, force, video, operator state) with per-modality calibration and an automated pipeline for episode segmentation and HDF5 export.

Vision-Based Proximity Estimation and Haptic Feedback for Surgical Teleoperation

Stereo perception pipeline on the da Vinci Research Kit, validated in a human-subjects study

- Built a synthetic data pipeline in Isaac Sim with domain randomization to fine-tune stereo disparity and semantic segmentation models in PyTorch for the dVRK stereo endoscope.
- Developed a real-time proximity detection pipeline from stereo endoscope images on the dVRK, producing per-pixel tool-to-environment distance estimates rendered to operators as haptic feedback.
- Validated the system in a 20-participant human-subjects study on the dVRK showing reduced operator cognitive load.

Selected Publications

1. S. Dalla Gasperina, D. H. Kang, H. Zhang, **A. F. Galvan**, et al. A bilateral teleoperation framework for dexterous manipulation. *IEEE RA-P*, 2026. In preparation.
2. **A. F. Galvan**, D. Rowland, A. Majewicz Fey. Toward vision-based proximity estimation and tactile feedback for surgical teleoperation. *IEEE RA-L*, 2026. To appear.
3. H. Boyea, **A. F. Galvan**, L. Sentis, A. Majewicz Fey. Povformer: Learning autonomous point-of-view control from teleoperated grasping and manipulation demonstrations. *IEEE RO-MAN*, 2026. To appear.
4. **A. F. Galvan**, A. Majewicz Fey. Characterizing human skill in surgical training using vision-derived contact. *IEEE RA-L*, 2026. Under review.
5. **A. F. Galvan**, et al. An extended virtual proxy haptic algorithm for dexterous manipulation in virtual environments. *IEEE WHC*, 2023. **Best Student Paper Award (Honorable Mention)**.